

Temporal Transcriptional Dynamics of Mesenchymal Progenitor-Like Cells During Axolotl

Limb Regeneration: A Single-Cell Re-Analysis Study

Executive Summary

The axolotl (*Ambystoma mexicanum*) has long been acknowledged as a dynamic biological model for dissecting complex tissue regeneration in vertebrates. Axolotls offer complete limb regeneration after amputation compared to mammals, reestablishing the bone, muscle, vasculature, and connective tissue without irreversible scarring. A major characteristic of this regenerative approach is the appearance of the blastema, which is a transient yet highly proliferative structure of progenitor-like cells, which eventually fills in the missing limb.

Recent studies through single-cell RNA sequencing (scRNA-seq) have reported cellular atlases of axolotl limb regeneration, yet relatively little interest has been paid to how the progenitor populations evolve over time. The aim of this work is a targeted re-analysis of publicly available scRNA-seq data on the temporality of transcriptional dynamics of mesenchymal progenitor-like cells in defined regenerative stages. This study aims to identify stage-specific gene expression programs for activation, proliferation, and tissue remodeling by comparing early injury (3h post-amputation and 24h post-amputation) and later regenerative stages (22 days and 33 days postamputation).

Outcomes for study include quantification of progenitor cell abundance over time and differentially expressed genes that characterize early vs late regeneration. Focusing on a biologically central cell population allows this work to shift from descriptive atlases and reach a mechanistic description of regeneration.

Literature Review

Regeneration in axolotls is highly coordinated and tightly regulated biological process. The axolotl passes through well-defined steps of regeneration after limb amputation, starting with fast closure of wounds and finishing with complete structural reconstruction. Unlike mammals and their normal tissue repair through fibrosis and scar formation, axolotls regenerate tissues in a coordinated and scar-free manner (Tanaka, 2016). The cellular and molecular basis of this difference remains a central question in regenerative biology.

Blastema is the key part of axolotl limb regeneration. Following closure of a wound, the cells below the apical epithelial cap are also activated and go back to the cell cycle creating a proliferative mass of progenitor-like cells (McCusker et al., 2015). These blastemal cells are not only undifferentiated stem cells, but are lineage-restricted progenitors retaining positional memory. This spatial identity enables cells to retain positional information along the proximal–distal axis (McCusker et al., 2015).

There are substantial contributions of this progenitor pool proposed from fibroblasts derived from connective tissue. These cells show outstanding plasticity, shifting from a differentiated state to an essentially proliferative progenitor-like state whilst reproducing spatial information. This plasticity and positional memory trade-off have been hypothesized to drive successful recovery.

At the genomic level, the axolotl genome harbors genomic regulatory elements responsible for tissue formation and regenerative regulation elements (Nowoshilow et al., 2018). Transcriptome-wide analyses have revealed genes associated with extracellular matrix remodeling, immune modulation, and cell cycle re-entry (Bryant et al., 2017). However, bulk

transcriptomic approaches average gene expression across heterogeneous populations, thereby obscuring cell type-specific dynamics.

Single-cell RNA sequencing has greatly expanded the field. scRNA-seq allows for precise assessment of individual cell states and the identification of rare or transitional populations. More recently, heterogeneous cell populations throughout limb regeneration have been mapped in recent datasets consisting of immune cells, epithelial populations, fibroblasts, and mesenchymal progenitor-like clusters (Li et al., 2021). Although these studies yield interesting cellular atlases, there is a lack of studies on cluster rather than stage-specific transcriptional transitions in established progenitor populations.

Since mesenchymal progenitor-like cells play a critical role in blastema development and tissue repair, investigating how this target population evolves in different stages of regeneration, they may provide valuable insights into regeneration-specific molecular programs.

Research Gap

While scRNA-seq studies have characterized cellular diversity after axolotl limb regeneration, time focused analysis of mesenchymal progenitor-like cells is limited. Existing work has largely focused on global cell-type annotation and paid little attention to how progenitor transcriptional programs shift from early injury to the later phase of remodeling.

Statement of the Problem:

The transcriptional programs that mediate mesenchymal progenitor-like cell behavior in early versus late stages of axolotl limb regeneration remain to be disentangled.

Significance

Mesenchymal progenitor-like cells are believed to be involved in blastema formation, structural patterning, and tissue reconstruction. And in explaining how these cells go through activation, proliferation, and remodeling stages, we can better understand why axolotls can successfully regenerate their cells, while mammals exhibit limited regenerative capacity.

More generally, characterizing stage-specific gene programming linked to regenerative competence might also be useful in the development of approaches to optimize tissue repair or minimize fibrosis in mammalian systems. By analyzing temporal dynamics in a central progenitor population, the mechanistic relevance, as well as the detailed descriptive analysis, can be given in this work.

Research Aims

1. Examine the proportional abundance of mesenchymal progenitor-like cells at various stages of regeneration.
2. Detect diverse expressions of genes associated with early (3h, 24h) vs late (22dpa, 33dpa) regeneration events in this group.
3. Determine biological correlates of early activation and late remodeling phases.

Proposed Methods / Approach.

Dataset.

The present study will leverage published integrated scRNA-seq data from axolotl limb regeneration (Li et al., 2021). It comprises an estimated 54,000 cells throughout 8 timepoints: control, 3h, 24h, 72h, 7dpa, 14dpa, 22dpa and 33dpa.

Computational Environment.

All analyses will be performed with R (version 4.3+) and Seurat (version 5) in a Conda-managed environment on a powerful computing cluster. Unified Seurat object includes normalized, scaled, batch-corrected expression data, so that later analyses are done from normalized states of cells.

Analytical Workflow.

We will first identify the mesenchymal progenitor-like cluster based on annotation and marker gene expression. This cluster subset will be divided for the focused temporal analysis.

It will then be further determined by computing the proportional fraction of this population by the timepoints. This step will decide if the mesenchymal progenitor-like cells show stage-dependent enrichment or depletion.

Third, cells will be divided into Early (3h and 24h) and Late (22dpa and 33dpa) stage of regeneration. This category captures biologically significant steps, which differentiate initial injury response/activation and subsequent structural remodeling and stabilization.

Differential gene expression analysis will be performed using the Seurat's

FindMarkers function, setting established thresholds for log fold change with adjusted p-value to achieve statistical rigor. Genes will be ranked according to magnitude of change or significance.

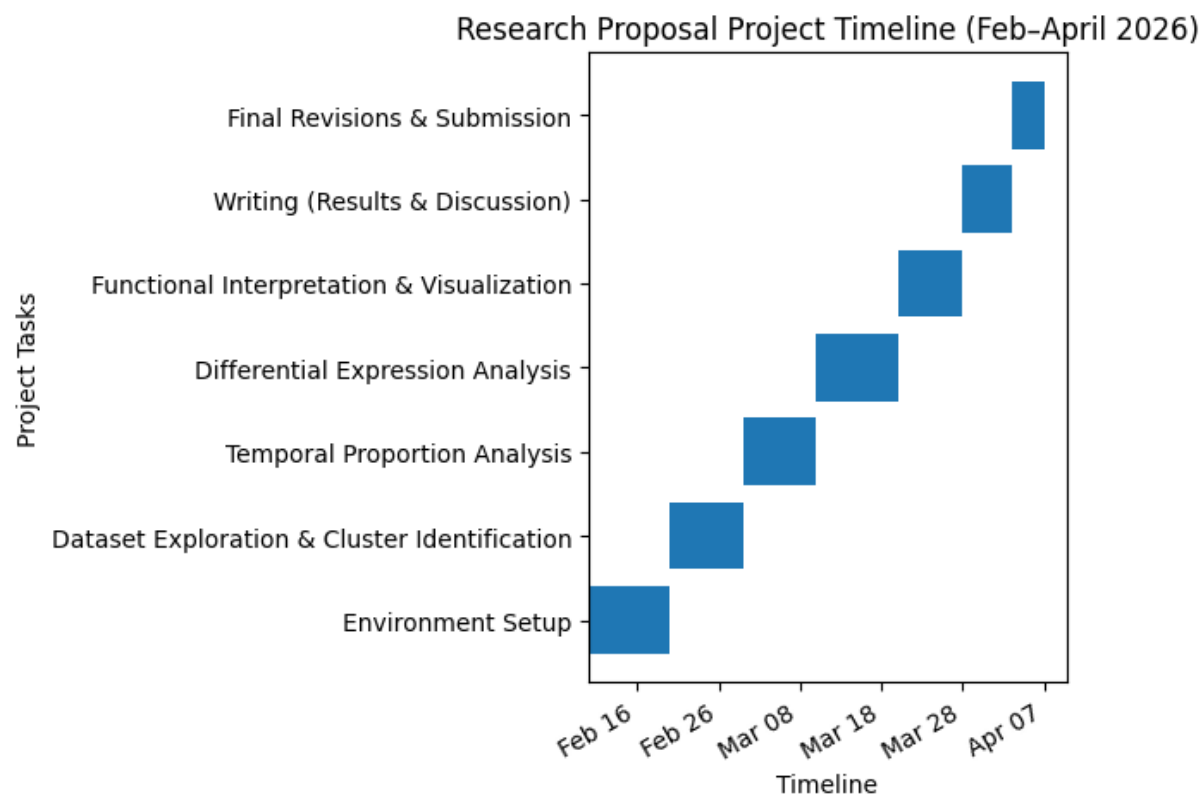
The visualization will include bar graphs showing proportionate abundance, volcano plots in which differential expression results are summarized, and heatmaps of top regulated genes.

When suitable, relevant genes interpreted from known pathways involved in proliferation, extracellular matrix remodeling, and differentiation.

Feasibility and Limitations.

The dataset has already been preprocessed and integrated and hence the proposed analysis, if possible, in time frame to execute, is computationally feasible. We have established the computing environment we will need, and initial exploration indicates that there are distinct progenitor-like populations across timepoints. Since this study involves a re-analysis of existing transcriptomic data, conclusions will be limited to inferring gene expression. To this end, no functional validation experiments are done, although experiments of certain performance can be useful as hypotheses for future experimentation.

Project Timeline



References

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